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Global soil organic carbon assessment

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A B S T R A C T

Soil carbon is a key component of functional ecosystems and crucial for food, soil, water and energy security. Climate change and altered land-use are having a great impact on soils. The influence of these factors creates a dynamic feedback between soil and the environment. There is a crucial need to evaluate the responses of soil to global environmental change at large spatial scales that occur along natural environmental gradients over decadal timescales. This work provides a suite of new data on global soil change which will uniquely utilize the world's prior investment in soil data infrastructure. Here we attempt a comprehensive global space–time assessment of soil carbon dynamics in different ecoregions of the world accounting for impacts of climate change and other environmental factors.

1. Introduction

By declaring 2015 as the International Year of Soils the United Nations General Assembly (A/RES/68/232) has recognized the major importance of the soil resource for human life and the need to maintain it for future generations. The intention underpinning this proclamation is to increase people's understanding about the important role of soil for food security through agricultural production, and the essential ecosystem functions that soil fulfils. In part, this has focussed the global agenda on understanding the effect land-use change has on soil properties and functions (Bouma and McBratney, 2013) as well as quantifying the relative impacts of climate change, including changes in temperature and rainfall patterns (Lal, 2004; Milly and Shmakin, 2002; Tao et al., 2003). To achieve this the current condition of global soil needs to be assessed (ITPS, 2015).

Understanding the biological, ecological, chemical and physical processes governing soil functions globally is directly related to most of the grand challenges in environmental science which have been outlined by the US National Academies of Sciences in 2010 (NRC, 2010). Because of the inherently long-term nature of soil change, addressing these questions requires empirical soil information collected over decades. In 1997, Trumbore (1997) recommended this course, i.e. to work on the evaluation of the response of soils to global environmental change at large scales that occur along natural environmental gradients over decadal timescales.

While there is a myriad of equally important soil indicators that could be used to gauge any substantial changes to soils, that may indicate greater potential degradation, the selection of such indicators needs to be: (i) acceptable to experts, (ii) routinely and widely measured, and (iii) have a currency with the broader population to achieve global acceptance and impact. The quantity of soil organic carbon (SOC) seems the obvious choice as it is arguably the most important soil indicator because of its central role in a range of soil functions, it is also one of the most common soil property measurements, and it could be argued that carbon itself is known to the global population (Karlen et al., 2001; Koch et al., 2013). More specifically, this choice stems from the known benefits of SOC for improved soil fertility and productivity and its contribution to food security which has been discussed widely within the *Soil Quality* concept (Andrews et al., 2004) and more

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recently within the wider *Soil Security* framework (McBratney et al., 2014a).

Over the past decades, there have been significant efforts to collect soil profile data around the globe to estimate the amount of SOC stocks worldwide. Such assessments have large uncertainties associated with them; for example Stockmann et al. (2013) discussed that global estimates of the size of the organic carbon pool to 1 m depth vary between 1463 and 2011 Gt C. The variation in these estimates could relate to measurements based on differing spatial data-sets, which consist of various sample sizes, but may also reflect data collected at different times. Moreover, these estimates only provide the total SOC stock at a particular time and do not indicate its temporal trend. A better understanding of global SOC estimates which incorporates an analysis of those factors that lead to changes over time is therefore warranted. Although efforts have been made globally to map the current state of soils or baseline levels of soil properties including SOC concentration, these have not reported any temporal trends (e.g. *GlobalSoilMap* project (Arrouays et al., 2014; Sanchez et al., 2009) and its offshoot *SoilGrids*1 *km* (Hengl et al., 2014) or the baseline SOC maps of Australia (Viscarra Rossel et al., 2014), Denmark (Adhikari et al., 2014), France (Meersmans et al., 2012) and the U.S. (Odgers et al., 2012)). Furthermore, any time-series soil carbon data are mostly available for a range of field trials; they are limited, however, as they only cover selected soils with defined management practices (e.g. Guo and Gifford, 2002; Sanderman and Baldock, 2010), or have been investigated at the national level (e.g. Bellamy et al., 2005; Chapman et al., 2013; Fantappiè et al., 2010; Reynolds et al., 2013). Currently, there is no global dataset that covers the whole range of spatio-temporal variation of environmental landscape conditions with different land uses and soil types.

In this work we gathered a suite of new soil profile 'time series' data on global soil change which will utilize the world's prior investment in soil data infrastructure assembled over the past half-century and beyond. Ultimately, our aim is to provide a comprehensive global space-time assessment of soil carbon dynamics for different biomes and ecozones of the world accounting for the impacts of climate change and other environmental factors. The analysis of these data could contribute to the design of a global SOC monitoring network, for ongoing measurement and provision of early-warning signals highlighting where the ability of the soil to perform a range of functions may be lost.

2. Global SOC assessment in space and time imperative for food security

Assessing SOC contents globally in space and time is of significance as this soil property is a vital indicator for evaluating soil condition. Over the years research has demonstrated that the maintenance of SOC concentrations is strongly linked to biological activity and agricultural productivity (Stockmann et al., 2013). Maintaining SOC contents above critical limits for specific ecological and climatic zones will help to protect soil resources and maintain crop yields thus contributing to global food security (Bouma and McBratney, 2013).

It has been suggested that where SOC concentrations are reduced below some critical limit, soil nutrient and water holding capacity will be impeded and physical degradation is likely to occur through fragmentation of soil aggregates and increased susceptibility to soil surface crusting and erosion (Amundson et al., 2015).

Various studies have found empirical critical values for SOC concentrations. A study by Aune and Lal (1997) showed that in tropical regions where SOC concentrations fall below 1.1%, crop yields are reduced by 20%. By contrast, Loveland and Webb (2003) reviewed the evidence for a critical value of 2% SOC for soils in temperate regions and reported that the quantitative evidence for such a threshold was slight, although there was evidence for a desirable range of SOC concentrations. Zvomuya et al. (2008) reported that a SOC content of 2% SOC was important for the productivity of soils. Work by Yan et al. (2000) identified a critical limit of SOC of 1.0% below which soil microbial diversity declined linearly. Hassink (1997) suggested that the capacity of soils to preserve SOC is dependent on its association with clay and fine-silt particles. Furthermore, Stockmann et al. (2013) proposed that a critical limit as well as an upper limit of SOC where SOC is saturated is related to the soil's texture (i.e. the amount of clay, silt and sand) in addition to other environmental factors. For example, based on the literature it was posited that a critical concentration of SOC (%) may be of $\sim 1.5k$ for sandy soils and $\sim 0.8k$ for clay rich soils, where k refers to a site or region-specific proportionality constant that depends on a variety of interacting environmental factors including climate and landuse. Such notional critical and saturation levels of SOC could have practical implications by guiding farmers to maintain sufficient organic matter in their soils to ensure optimum productivity is maintained.

Research cited above confirms the necessity for identifying potential SOC deficits worldwide. Once the world soils fall under a certain critical limit, and once this tipping point is crossed, the soil resource has low resilience to recover and accumulate SOC to previous levels. There are a number of events in history that have demonstrated that misuse of the soil can lead to the destruction of the soil's fertility which can impact immensely on society. The 'dust bowl' that occurred in the 1930s in the US and Canadian prairies is such an example which demonstrates that the inappropriate use of agricultural practices can lead to severe agricultural damages and ecosystem failures (Montgomery, 2007a, 2007b). Studying the feedbacks between global and regional climate on the amount of terrestrial SOC including the role of land-use change is therefore of great importance (NRC, 2010).

Recently, Carvalhais et al. (2014) proposed a mean residence time of soil organic carbon of 23 years based on global soil profile data from the Harmonized World Soil Database. The authors took a simplistic approach by calculating the mean turnover time of C(T) through the ratio between carbon stored in vegetation (C_{plant}) and soils (C_{soils}), and the flux into this C storage (GPP):

$$T = \frac{(C_{plant} + C_{soil})}{GPP} \qquad (1)$$

3. A global space–time assessment of SOC

Assessing the world's SOC concentrations spatially and temporally comes with a number of challenges, most of which will be addressed in the following.

3.1. Data collation for mapping SOC change

Within this work we focused on the collation and subsequent synthesisation of global historical and recent SOC profile data, sourced from regional, nationwide or global soil (test) databases (Table 1), including the time and depth of sampling, geographical location and analytical methods used.

In addition, where available we acquired additional basic soil property data, most importantly clay content and bulk density as this information is crucial to ultimately assess changes in SOC over time based on an equivalent mass approach (Ellert and Bettany, 1995; Ravindranath and Ostwald, 2008). Although very important,

Table 1
Soil profile data sources.

Data from ISRIC (World Soil Information) www.isric.org:
ISRIC-WISE-Global Harmonized Soil Profile Data
ISRIC-WISE international soil profile dataset
Africa Soil Profile Database
Soil and terrain database (SOTER) for Central and Eastern Europe
Soil profiles from the SOTER China
Soil and terrain database for Latin America and the Caribbean (SOTERLAC)
NCSS (National Cooperative Soil Survey-National Cooperative Soil Characterization Database, online at http://ncsslabdatamart.sc.egov.usda.gov)
NASIS Soil profile database (United States Department of Agriculture-Natural Resource Conservation Service (USDA-NRCS))
Agriculture and Agri-Food Canada
Soil Pedon Database – CanSIS
Oak Ridge National Laboratory (ORNL) Distributed Active Archive Center (DAAC) http://daac.ornl.gov
Soil Profiles of Amazonia – Pre-LBA RADAMBRASIL Project Data
Soil Properties of Forested Headwater Catchments, Mato Grosso, Brazil – LBA-ECO ND-11
Plant and Soil C and N Isotopes, Southern Africa 1995-2000 – SAFARI 2000
Soil Composition and Structure in the Brazilian Amazon 1992-1995 – LBA-ECO LC-09
Forest and Pasture Soil and Grass Analyses, Rondonia, Brazil 2003-2004 – LBA-ECO ND-01
Soil Lab Data – BOREAS TE-01 SSA
Zinke et al. (1986) Global Organic Soil Carbon and Nitrogen Data Set. http://www.daac.ornl.gov
Instituto Nacional de Estadística, Geografía e Informática (INEGI) Conabio Mexico
Soil Profile database for Mexico
Centro de Información de Recursos Naturales (CIREN) Chile
Soil Profile data of Chile
Rural Development Agency (RDA) Korea
Soil Profile database for South Korea
European Soil Portal, Joint Research Centre, European Commission http://eusoils.jrc.ec.europa.eu
Land Use/Cover Area frame Statistical Survey – LUCAS Topsoil dataset
The Soil Profile Analytical Database for Europe – SPADE/M
British Geological Survey and the Geological Survey of Northern Ireland
Soil Profile database for Northern Ireland
Landcare research New Zealand
Land Resource Information (LRIS) dataset
Australia
National Soil Data Collation, Terrestrial Ecosystem Research Network (TERN) Soils Project
CSIRO Land and Water Flagship
The Northern Territory Department of Land Resource Management
The Queensland Department of Science, Information Technology, Innovation and the Arts
The South Australia Department of Environment, Water and Natural Resources
The Tasmania Department of Primary Industries, Parks, Water and Environment
The Victoria Department of Environment and Primary Industries
The Western Australia Department of Agriculture and Food
Geoscience Australia
The Australian Collaborative Land Evaluation Program
Department of Environmental Sciences, The University of Sydney

inorganic C data were not readily available globally and have therefore not been included in the assessment of global SOC concentrations. We collected a total of 94,381 soil profiles around the world, with most of them covering soil information of up to one meter depth. Of these, around two thirds (63,503) have a definite sampling date (Fig. 1).

The importance of soil legacy data for digital soil mapping has been demonstrated in various studies over the years. In general terms, first they are an important data source for spatial mapping exercises, and secondly they can be a means to detect temporal changes in soil properties. For example, Lindert et al. (1996a, 1·996b) used soil survey data (1930–1980) from southern and northern China and showed temporal trends in soil chemical characteristics in relation to changes in cropping intensity and fertilizer application. Furthermore, work from France has demonstrated that a nationwide soil test database, based on routine analysis of soil samples provided annually by the nation's farmers, can be used to detect decadal spatiotemporal changes in SOC (Angers et al., 2011; Saby et al., 2008) and phosphorus (Lemercier et al., 2008). Similarly, Fantappiè et al. (2010) estimated the decadal variability of SOC stocks in Italy employing a national soil database (1990–2010) and found that changes in SOC were not only controlled by land use, soil and environmental variables but also by the period of observation, with SOC stocks decreasing notably between the first and second decade but in turn also increasing again in the third decade, from a total of 3.32–2.74 Gt and 2.93 Gt, respectively. van Wesemael et al. (2010) conducted an inventory of Belgian soils dating from the 1960s and 2006 and found that agricultural management practices and land-use impacts accounted for more than 70% of the variation in observed SOC changes between the years. Minasny et al. (2011) utilized soil legacy data collected at various time periods (1930–2010) from Java, Indonesia, to investigate soil organic carbon dynamics in relation to land use change. Their research showed that SOC stocks in Java increased starting from the 1970s after changes in management practices, with a net-accumulation rate of about 0.2–0.3 $tCha^{-1}\,yr^{-1}$ in the top 10 cm during the period of 1990–2000. Similarly, Minasny et al. (2012) found an increase in measured SOC equivalent to about 0.3 Tg (10^{12} g) C yr^{-1} (1970–2010) for the soils of South Korea due to an increase in above- and below- ground biomass due to fertilization and good agronomic practice. More recently, Deng et al. (2013) were able to track temporal changes in SOC of Danish soils using near-infrared spectroscopy, whereas Kurganova et al. (2014) identified the positive feedback for SOC accumulation after the collective farming collapse and abandonment of agricultural land in Russia (1990–2009), with an annual sequestration rate of 42.6 Tg C yr^{-1}.

3.2. Data standardization for mapping SOC change

Utilizing historical and recent data for the mapping of SOC change however introduces the challenge of data harmonization (Sulaeman et al., 2013). For the digital soil mapping approach, SOC soil profile data have therefore been standardized in terms of analytical methods, depth and time. The Walkley-Black method was commonly used to measure SOC worldwide but has recently been replaced with dry combustion or even spectroscopic methods such as MIR or VIS–NIR. Pedotransfer functions are therefore used to harmonize SOC data to a common measuring method (Skjemstad et al., 2000). SOC profile data are also standardized to fixed depths following the *GlobalSoilMap.net* specifications (0, 5, 15, 30, 60, 100 cm), employing the equal-area spline approach (Malone et al., 2009). We assess SOC levels for the 60 s, 80 s, 90 s, 2000 s, however in this work we will show results for the 2000s only. The collection of SOC data at various points in time was accounted for by introducing the 4th dimension into our modeling efforts. There is however a 'space for time' trade-off within our global dataset which needs to be accounted for. More specifically, data points do not have equal spatial coverage through time.

3.3. Spatial modeling and modeling of temporal change of SOC dynamics

We used the digital soil mapping framework of the *scorpan* regression kriging approach (McBratney et al., 2003) to model the spatial and temporal distribution of soil carbon around the globe. The scorpan approach involves the assembling of a set of global environmental covariates that are chosen to represent a range of soil forming factors and drivers of soil change, including digital elevation models and their derivatives, climate data (long-term precipitation and temperature data), and landcover data. Landcover was identified as one of the major drivers of temporal SOC

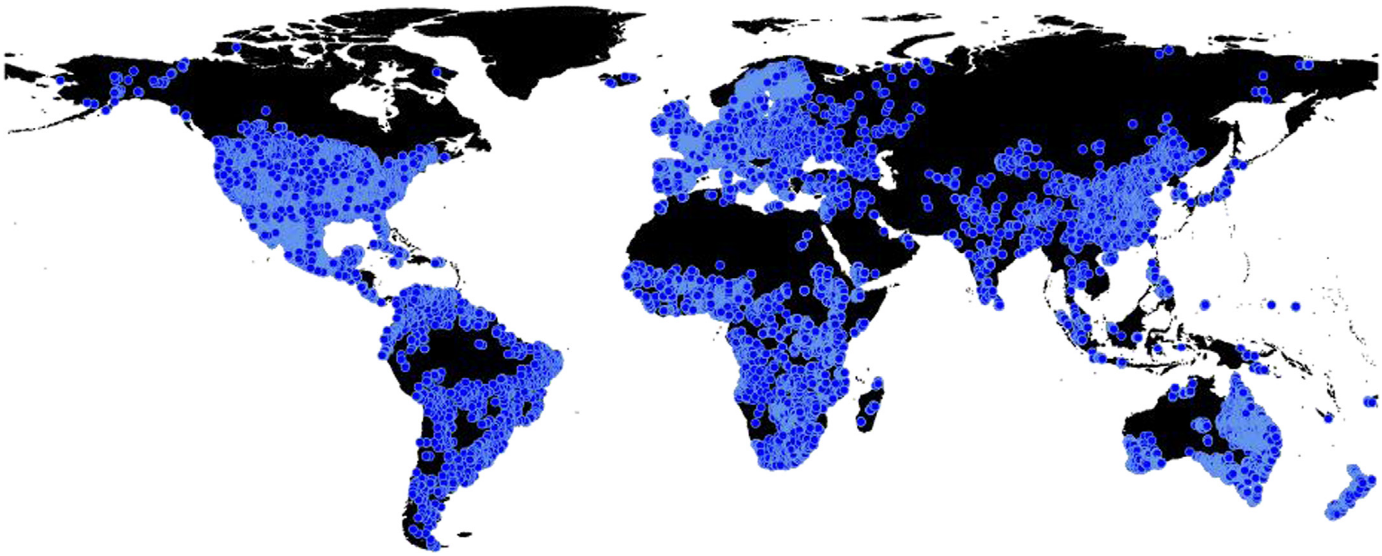

Fig. 1. Location of soil profiles used for global spatial and temporal SOC assessment that are associated with a definite date of observation (about 63,503 soil profiles).

change during modeling preparations. Temporal modeling efforts were therefore conducted as a function of landuse, with landcover classifications being extracted for each year at the soil profile data sampling locations. Data mining tools were then applied to extract soil spatial prediction functions, followed by kriging of the residuals, and the determination of the uncertainty of the predictions (Malone et al., 2011) for each depth increment and time frame. This is achieved initially at a spatial resolution of 1 km, with the option of increasing the resolution to 100 m, employing downscaling using the method described in Malone et al. (2012). Data analysis and modeling efforts were conducted employing the *Google Earth Engine* platform (Hansen et al., 2013; Padarian et al., 2015).

In the following section, we present topsoil (0–10 cm) SOC content (%) for the year 2001 as compared to 2009. For this time period global landcover change has been analysed using the MODIS Land Cover Type product (MCD12Q1) generated by the Land Processes Distributed Active Archive Center, U.S. Department of the Interior and U.S. Geological Survey (http://lpdaac.usgs.gov/products/modis_products_table/mcd12q1).

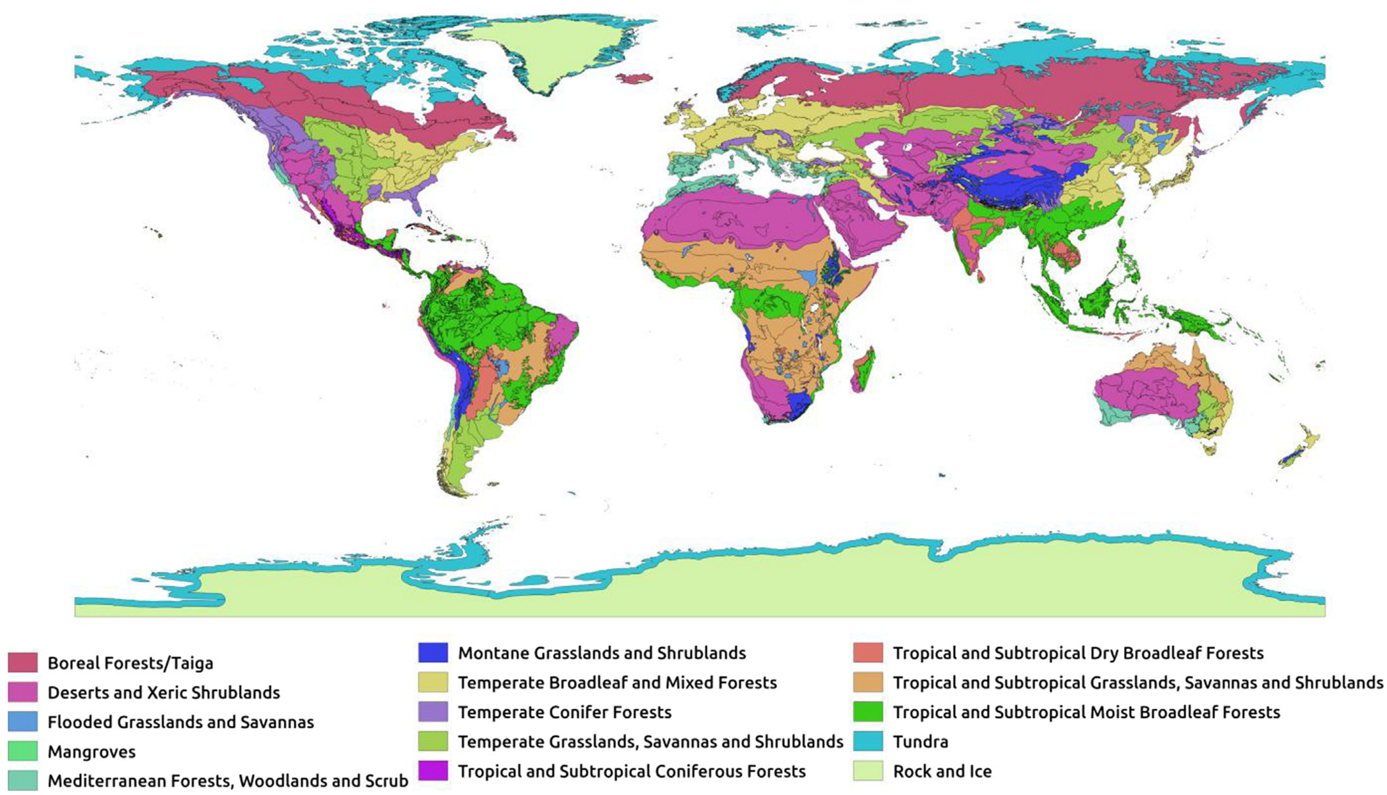

Fig. 2. Map showing Global Ecoregions used in this work. Global Ecoregion coverage is sourced from The Nature Conservancy, USDA Forest Service and U.S. Geological Survey, ecoregional assessments (metadata can be found at http://maps.tcn.org/files/metadata/TerrEcos.xml.).

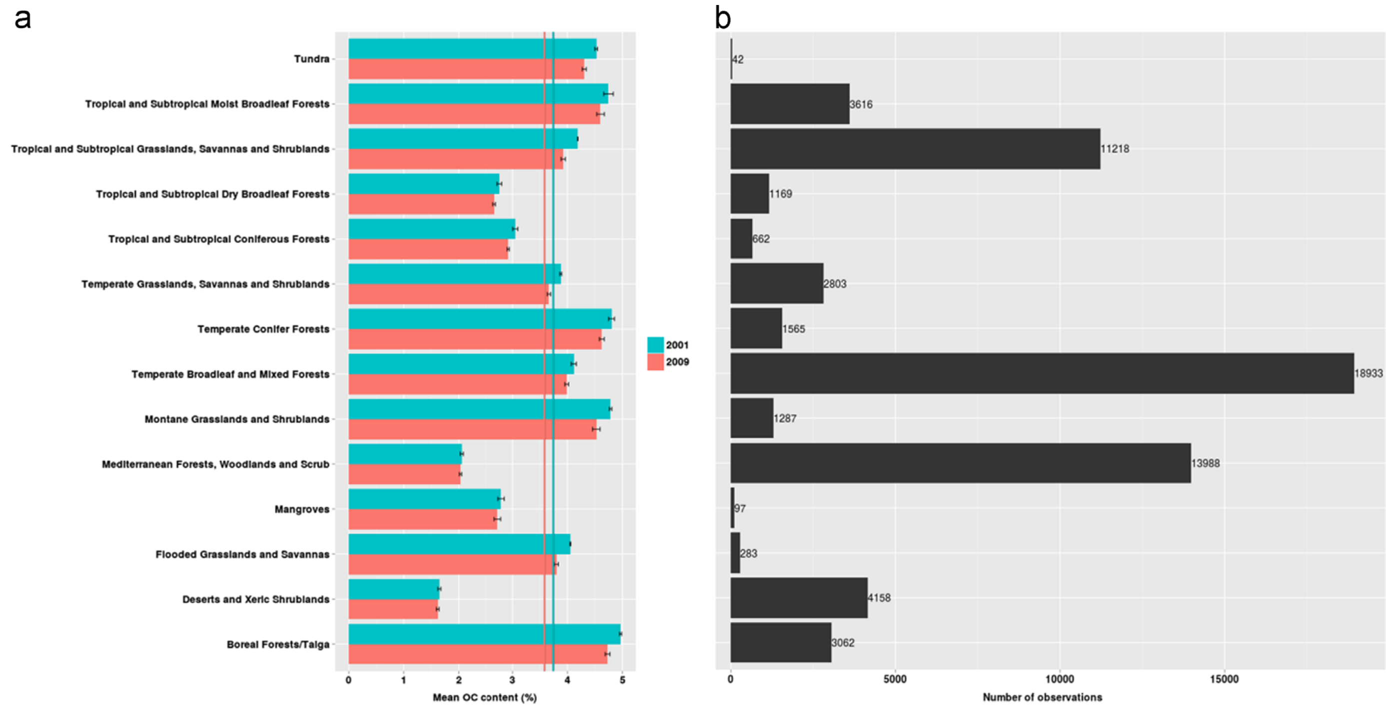

Fig. 3. (a) Mean topsoil SOC content (0–10 cm) by Global Ecoregions for the observation year 2001 as compared to 2009. Global Ecoregion coverage is sourced from The Nature Conservancy, USDA Forest Service and U.S. Geological Survey, ecoregional assessments (metadata can be found at http://maps.tcn.org/files/metadata/TerrEcos.xml.) Black bars represent standard deviation of the iterations. Values of mean global SOC contents for 2001 and 2009 of 3.94% (green line) and 3.76% (red line) respectively, are also shown. (b) Soil profile distribution per ecoregion per year. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4. Global topsoil SOC change for the benchmark of the 2000s

Here, we show first results from our global SOC assessment efforts in space and time.

4.1. Mean topsoil SOC change for global ecoregions for the observation years 2001 and 2009

Fig. 2 shows mean topsoil SOC contents (0–10 cm) for the ecoregions of the world for the observation year 2001 as compared to 2009. Global Ecoregion coverage (Fig. 3) is sourced from The Nature Conservancy, USDA Forest Service and U.S. Geological Survey, ecoregional assessment (metadata can be found at http://maps.tcn.org/files/metadata/TerrEcos.xml). These topsoil SOC concentrations are calculated from averages of the spatio-temporal model over those regions not simply from the datapoints themselves.

In Fig. 3, benchmark values of mean global SOC contents for

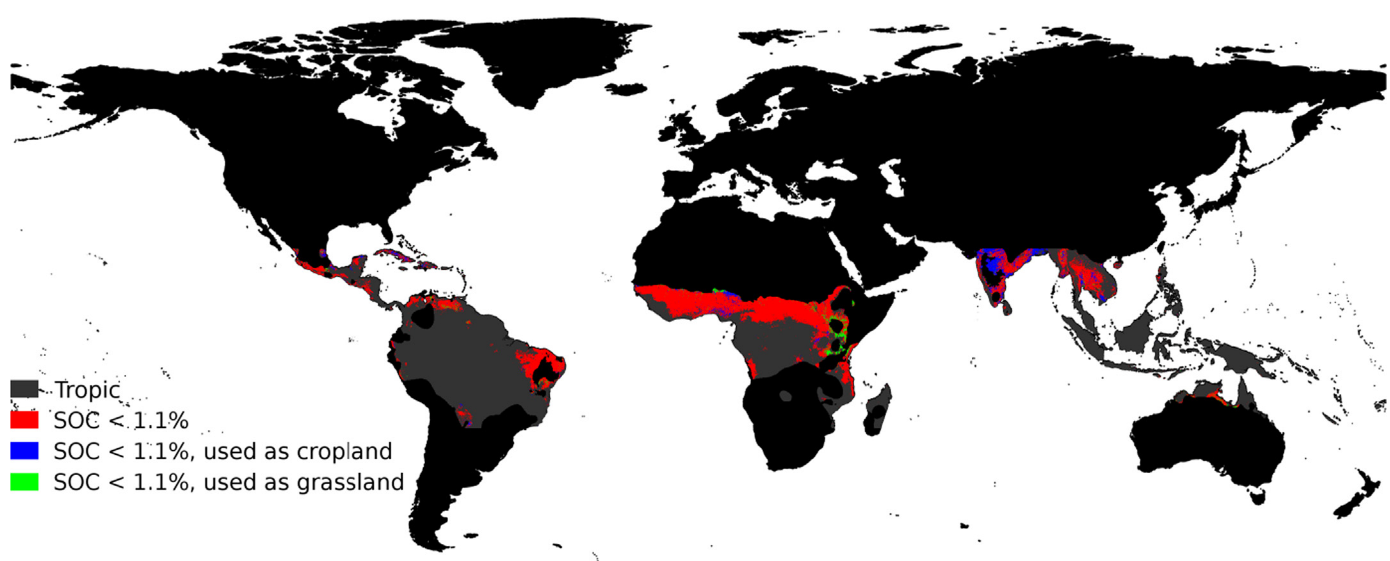

Fig. 4. Map of soil carbon content where soils of the Tropics fall below a soil organic carbon content of 1.1% for the observation year of 2009. Of these, areas that are used as cropland are shown in blue and areas that are used as grassland are shown in green. The Tropics were classified using the world map of the Köppen–Geiger climate classification (Peel et al., 2007). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

2001 and 2009 of 3.94 ± 0.03% and 3.76 ± 0.03% respectively, are also shown as well as the soil profile distribution per ecoregion per year used in the modeling efforts which can be used as an indicator of uncertainty of the predictions. In general, our model predicts that most ecoregions have experienced a loss of SOC in 2009 as compared to 2001 or have not experienced a change in mean SOC. These changes, however, need to be investigated further as SOC change can also be related to errors in the predictions related to misclassifications of landcover from the MODIS product.

4.2. Mean topsoil SOC change for the Tropics for the observation years 2001 and 2009

As discussed in Section 2, various authors have found varying empirical levels of critical SOC contents below which crop yields show noticeable decline. For the area of the Tropics, Aune and Lal (1997) proposed a cut-off of 1.1% SOC. In this regard Fig. 4 shows areas of the Tropics where SOC contents fall below the proposed concentration of 1.1% of SOC (indicated in red) for 2009. Areas under some form of management important for food production, i.e. cropland (indicated in blue) and grassland (indicated in green) that are below critical levels are also shown. However, here, we need to consider that the soil's texture has not been taken into account. Most likely, for some of the regions identified of being at risk that are associated with a relatively high clay content, cut-off values would be less than 1.1% because of the need of region-specific adjustment of empirical critical values (also refer to discussions in Section 2 and McBratney et al., 2014b). Therefore, in the future, global clay contents need to be considered to clearly identify regions of yield limiting SOC levels.

Fig. 5a shows a region of the Tropics, the State of Pará, Brazil, where SOC contents have experienced a loss of up to 1% between 2001 and 2009 (areas shown in red). As can be seen in Fig. 5b and c, a loss in SOC is also related to changes in land use. In the region shown here, it is clearly visible that areas that are used for cropland in 2009, are still areas of natural vegetation in 2001, SOC loss is therefore most likely related to changes in management.

We need to note here, however, that changes in SOC predicted by our model are to a point driven by land use change, i.e. in areas where land use change occurred between 2001 and 2009, the model predicts that SOC contents have changed, positively or negatively. This clearly needs to be investigated further.

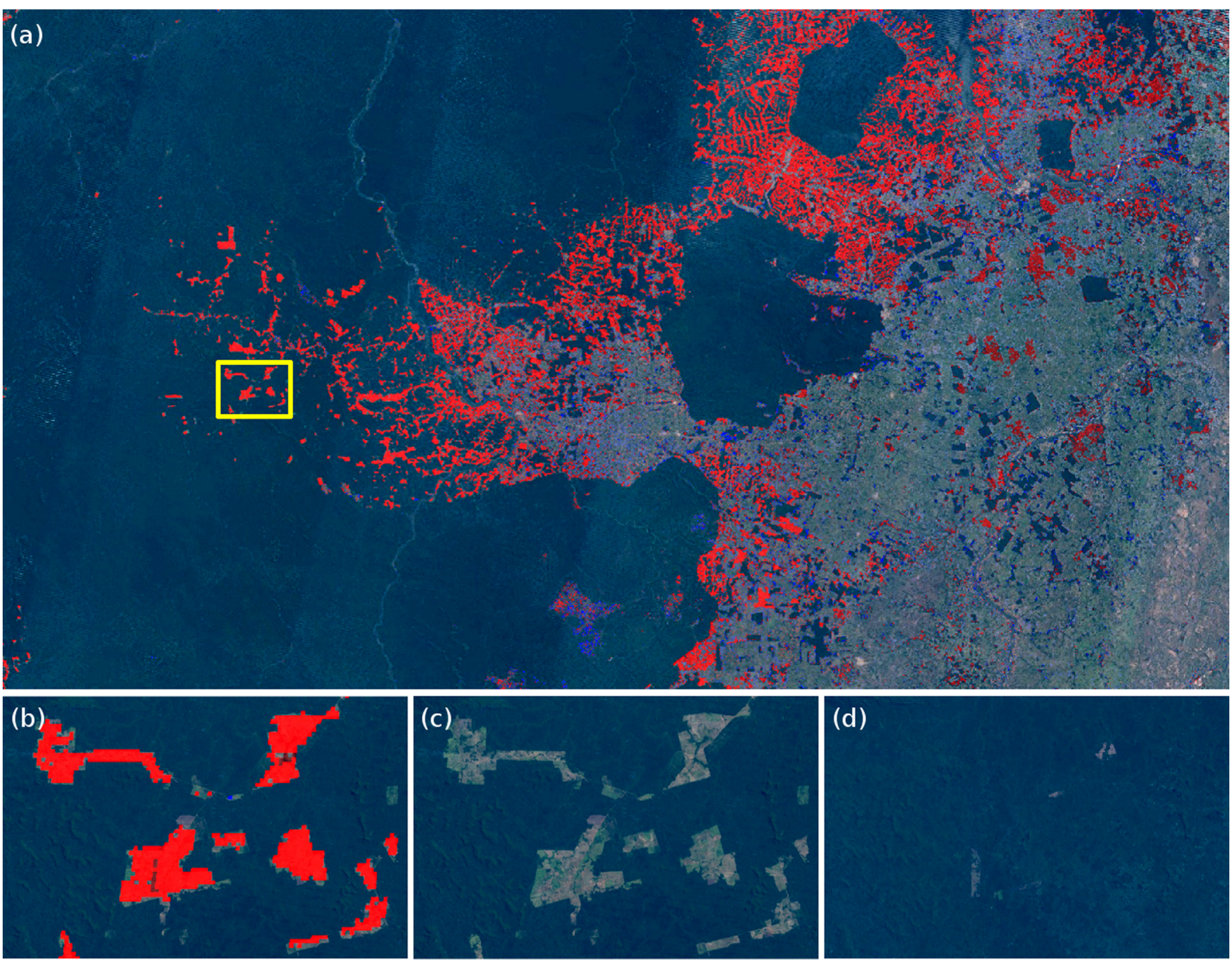

Fig. 5. (a) Topsoil SOC loss indicated in red (i.e. a loss of up to 1% SOC) for the observation year of 2009 as compared to 2001 for the State of Pará, Brazil. (b) Close-up of an area indicating soil loss for the year 2009 as compared to 2001. (c) Satellite image for the year 2009. (d) Satellite image for the year 2001. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5. Conclusions and future work

This work provides a unique and new global assessment of SOC based on spatial and temporal data that will result in a number of maps showing the worldwide change in SOC over time. Main drivers, i.e. those that have the most impact on the temporal change of SOC will be identified for a range of regions worldwide. Preliminary results showed that landcover change is the primary agent that influences SOC change over time, followed by temperature and precipitation. Questions such as where in the world is SOC declining or growing will be answered through the final product of this work.

Moreover, tracking the changes in SOC through time will help to identify where our world soils are close to reaching critical thresholds of sustainability and will thus help to identify places under which environmental conditions are likely to provoke maximal change. Diagnosing the condition of the soil resource using SOC as a proxy will ultimately help to prevent threating processes, such as a loss of soil structural integrity and soil biodiversity, as well as a decline in soil fertility which will in turn secure sustainable food production.

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